

A Capacity-Radius List-Decoding Obstruction for the $m=1$ Specialization of Reed-Solomon Codes

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Front Matter

This draft claims a partial result: resolution of the $m=1$ specialization of the grand Reed-Solomon list-decoding challenge at the capacity-radius surface. It does not claim to resolve the full mutual correlated agreement challenge or the full grand list-decoding challenge for arbitrary constant m .

GitHub timestamp repository: <https://github.com/thehenryco/rs-m1-capacity-list-obstruction>
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Abstract

We give a capacity-radius list-decoding obstruction for the $m=1$ specialization of Reed-Solomon codes. Let $C = \text{RS}[F_q, L, k]$ with $|L|=n$. We prove that, whenever $q > \text{binom}(n, k+1)$, there exists a received word y for which the Hamming ball of radius $n-k$ contains exactly $\text{binom}(n, k)$ Reed-Solomon codewords. Consequently, if $\text{binom}(n, k+1) < q < 2^{128} \text{binom}(n, k)$, then $|\text{Lambda}(C, y, n-k)| = \text{binom}(n, k) > 2^{-128} |F_q|$. Thus the $\epsilon_{\text{star}} = 2^{-128}$ threshold fails at $\delta = 1-k/n$ for the $m=1$ specialization in this field-size window.

1. Scope of the Claim

This draft addresses the $m=1$ specialization of the grand list-decoding challenge. In this specialization, $\text{Lambda}(C, \delta)$ is the ordinary Reed-Solomon list at relative Hamming radius δ . The draft does not assert a result for arbitrary constant m unless an explicit m -wise extension is added.

2. Notation

Let F_q be a finite field and let $L \subset F_q$ be an evaluation domain of size n . Let $\text{RS}[F_q, L, k]$ denote the Reed-Solomon code obtained by evaluating polynomials of degree less than k on L . The rate is $\rho = k/n$. For a received word y in F_q^n and integer radius r , let $\text{Lambda}(C, y, r)$ be the set of codewords of C within Hamming distance r from y .

3. Main Theorem

Let $C = RS[F_q, L, k]$ with $|L|=n$. Suppose $q > \text{binom}(n, k+1)$. Then there exists y in F_q^n such that $|\text{Lambda}(C, y, n-k)| = \text{binom}(n, k)$. Consequently, if $q < 2^{128} \text{binom}(n, k)$, then $|\text{Lambda}(C, y, n-k)| > 2^{128} |F_q|$ at $\delta = 1-k/n$.

4. Proof

Choose y in F_q^n such that no polynomial of degree less than k agrees with y on more than k coordinates. Section 5 proves that such y exists when $q > \text{binom}(n, k+1)$.

For each k -subset S of the n coordinate positions, there is a unique degree $< k$ polynomial f_S interpolating y on S . Its Reed-Solomon codeword agrees with y on S , hence has Hamming distance at most $n-k$ from y .

If two distinct k -subsets S and T produced the same polynomial, then that polynomial would agree with y on $S \cup T$. Since S and T are distinct k -subsets, $S \cup T$ has more than k coordinates. This contradicts the choice of y . Therefore the $\text{binom}(n, k)$ k -subsets produce $\text{binom}(n, k)$ distinct codewords in $\text{Lambda}(C, y, n-k)$.

Conversely, any codeword in $\text{Lambda}(C, y, n-k)$ agrees with y on at least k coordinates. By construction, no degree $< k$ polynomial agrees with y on more than k coordinates. Therefore each listed codeword agrees on exactly k coordinates and is generated by one of the k -subsets above. Hence $|\text{Lambda}(C, y, n-k)| = \text{binom}(n, k)$.

If $q < 2^{128} \text{binom}(n, k)$, then $\text{binom}(n, k) > 2^{128} q = 2^{128} |F_q|$. Thus the threshold $|\text{Lambda}| \leq 2^{128} |F_q|$ fails at $\delta = 1-k/n$.

5. Existence Lemma

Choose y uniformly at random from F_q^n . Fix a degree $< k$ polynomial f and a fixed $(k+1)$ -subset T of coordinates. The probability that f agrees with y on every coordinate of T is $q^{-(k+1)}$. There are q^k degree $< k$ polynomials and $\text{binom}(n, k+1)$ choices for T . By the union bound, the probability that some degree $< k$ polynomial agrees with y on more than k coordinates is at most $\text{binom}(n, k+1)/q$. If $q > \text{binom}(n, k+1)$, this probability is less than 1, so a suitable y exists.

6. Fixed-Rate Examples

$\rho = 1/2$

- $n = 32$
- $k = 16$
- $\delta = 16/32$
- radius = 16
- exact list size = 601080390
- existence threshold: $q > 565722720$
- threshold failure condition: $q < 2^{128} * 601080390$

$\rho = 1/4$

- $n = 32$
- $k = 8$
- $\delta = 24/32$
- radius = 24
- exact list size = 10518300
- existence threshold: $q > 28048800$
- threshold failure condition: $q < 2^{128} * 10518300$

$\rho = 1/8$

- $n = 32$
- $k = 4$
- $\delta = 28/32$
- radius = 28
- exact list size = 35960
- existence threshold: $q > 201376$
- threshold failure condition: $q < 2^{128} * 35960$

$\rho = 1/16$

- $n = 32$
- $k = 2$
- $\delta = 30/32$
- radius = 30
- exact list size = 496
- existence threshold: $q > 4960$
- threshold failure condition: $q < 2^{128} * 496$

7. Prior-Work Positioning

. The final version should compare this result with the Proximity Prize companion paper, classical Reed-Solomon list decoding, Johnson-type bounds, capacity-radius behavior, and known impossibility or near-capacity results. This section must identify whether the $m=1$ specialization is explicitly covered by the prize definitions or whether it is submitted as a specialization/partial result.

8. Submission Readiness

This document is a pre-submission draft until the following are added: final PDF, public repository timestamp, peer-review acceptance information, expanded prior-work comparison, and final conflict disclosure.

Public Timestamp and Repository Status

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Prior-Work Positioning

The Proximity Prize targets Reed-Solomon list-decoding and correlated-agreement challenges formalized in the companion paper by Arnon, Boneh, and Fenzi. This draft is positioned as an $m=1$ ordinary-list specialization, not as a claim about the full mutual correlated agreement object or arbitrary constant m .

Classical Reed-Solomon list decoding studies the size of Hamming balls around received words and the radius at which many codewords can appear. The construction in this draft is elementary but directly exposes the capacity-radius surface: for $C = \text{RS}[\mathbb{F}_q, L, k]$, a received word y with no degree $< k$ polynomial agreeing on more than k coordinates has exactly $\text{binom}(n, k)$ codewords at radius $n-k$.

Johnson-type bounds and capacity-style results are usually used to upper-bound list size below or near decoding thresholds. This note instead gives a lower-bound obstruction at the capacity-radius boundary $\delta = 1-k/n$. It should therefore be read as a specialization/edge-case obstruction that must be compared carefully against the exact $\Lambda(C^{\{\equiv m\}}, \delta)$ definitions in the Proximity Prize companion paper.

Recent work around Reed-Solomon proximity gaps relates proximity-gap behavior to list-decodability limits. This draft does not claim to improve those results. It isolates the $m=1$ ordinary-list object and supplies a transparent construction showing threshold failure in the field-size window $\text{binom}(n, k+1) < q < 2^{128} \text{binom}(n, k)$.

Companion-Paper Notation Alignment

This draft aligns to the $m=1$ specialization of the grand list-decoding notation. In that specialization, $C^{\{\equiv 1\}}$ is treated as the ordinary Reed-Solomon code C , and $\Lambda(C^{\{\equiv 1\}}, \delta)$ is treated as the ordinary list of codewords within relative Hamming radius δ of a received word y .

The theorem proves that at $\delta = 1-k/n$, there exists y such that $|\Lambda(C^{\{\equiv 1\}}, y, \delta)| = \text{binom}(n, k)$. When $\text{binom}(n, k+1) < q < 2^{128} \text{binom}(n, k)$, this exceeds $2^{-128} |\mathbb{F}_q|$.

This notation alignment does not assert the arbitrary constant- m case. The arbitrary- m object remains outside this draft unless a separate m -wise construction is added.

Final Conflict Disclosure

The author declares no known financial, institutional, employment, advisory, or personal conflicts of interest that would affect this pre-submission draft. The public GitHub repository is owned by the author through the thehenryco GitHub account.

Academic Repository Metadata

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Suggested arXiv categories: cs.IT, math.IT, cs.CR

Suggested IACR area: coding-theory-adjacent foundations for proximity testing and list decoding.